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Lab Report - Group 5 - Experimental Fluid Dynamics

# Wind Tunnel: Pressure and Velocity Measurements

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## 1. Introduction

This report presents the findings from a wind tunnel experiment conducted as part of the Experimental Fluid Dynamics course. The study focuses on the aerodynamic characterization of a NACA 23012 airfoil, a commonly used profile in aerospace applications. The primary objectives are to measure the pressure distribution across the airfoil's surface, calculate the lift ($C_L$) and drag ($C_D$) coefficients, and analyze the wake dynamics behind the airfoil to estimate total drag. These results are used to construct the $C_L$ vs. $\alpha$ and polar curves, which are essential for understanding the airfoil's performance across a range of angles of attack.

The experiment is divided into two distinct phases. In the first phase, some pressure transducers embedded in the airfoil surface record pressure distributions at various angles of attack. The collected data is used to determine the aerodynamic forces acting on the airfoil and derive the corresponding lift and drag coefficients.

In the second phase, a hot-wire anemometer is used to measure the velocity field in the wake of the airfoil. By moving the sensor vertically at different heights, the wake velocity profile is mapped to estimate total drag.

The combination of these measurements allow for an alternative way to compute forces on the airfoil, besides the direct force measurement shown in the previous laboratory report.

## 2. Experiment Description and Setup

The experiments were conducted in a calibrated wind tunnel designed to provide steady and controlled airflow conditions. The objective was to characterize the aerodynamic performance of a NACA 23012 airfoil by analyzing the pressure distribution across its surface and the velocity field in its wake. The experiment was divided into two main phases: pressure distribution measurements and wake velocity field analysis.

### 2.1. Equipment and Measurements

- **Wind Tunnel:** The wind tunnel generates a uniform and steady flow with a maximum freestream velocity of approximately 21 m/s. This is equivalent to a Reynolds number of approximately $1.5 \cdot 10^5$. The test section is transparent and some ports for pressure and velocity measurements.
- **Airfoil:** A NACA 23012 airfoil was selected. The airfoil was fitted with 16 pressure transducers along its surface to capture the pressure distribution accurately (8 in the upper-surface and 8 in the lower-

surface). These measurements were essential for calculating the lift coefficient ($C_L$) and the drag coefficient ($C_D$), as well as generating the airfoil's polar curve.

- **Reference Pressure Transducers:** Two additional pressure transducers were placed upstream of the airfoil to measure reference pressures. These reference values were used to normalize the pressure readings on the airfoil's surface and compute the dynamic pressure in the test section.
- **Hot-Wire Anemometer:** A single-wire hot-wire anemometer was used to measure the velocity field in the wake of the airfoil. This instrument provided precise velocity measurements by moving vertically in the $Z$ direction to map the wake velocity profile at various heights. The hot-wire sensor was calibrated prior to the experiment to ensure accuracy.

## 2.2. Experimental Procedure

The experiment consisted of two main phases:

### 2.2.1 Pressure Distribution Measurements

1. The airfoil was mounted in the test section and adjusted to a range of angles of attack, from $-12°$ to $20°$, in increments of approximately $2°$.
2. The pressure transducers embedded in the airfoil's surface recorded the pressure distribution for each angle of attack. These pressures are measured differentially, taking the pressure at the outlet of the convergent section - $p_2$- as a reference.
3. The lift coefficient ($C_L$) and pressure drag coefficient ($C_D$) were calculated by integrating the pressure distribution using MATLAB, which is explained below. The results were used to generate the $C_L$ vs. $\alpha$ curve and the airfoil's polar plot.

### 2.2.2 Wake Velocity Field Analysis

1. After completing the pressure distribution measurements, the hot-wire anemometer was calibrated using a known velocity profile in the wind tunnel.
2. The airfoil was tested at a specific angle of attack, chosen to analyze wake characteristics in greater detail.
3. The hot-wire sensor was moved vertically across the wake in the $Z$ direction, recording velocity values at various heights. This allowed for the construction of a velocity field map and the identification of the wake region behind the airfoil.
4. The wake velocity data was used to estimate the total drag coefficient ($C_D$) by applying a momentum deficit approach.

# 3. Experiment theoretical background

The ultimate goal of these experiments is to convert classical measurements of pressure and velocity into values for the lift and drag provided by the airfoil in different angles of attack. Ultimately, the goal is to reproduce the usual $C_L$ vs $\alpha$ and polar curves for the given airfoil. In order to do so, it is necessary to apply some theoretical developments.

On one hand, it is possible to compute the pressure forces exerted on the airfoil by integrating the experimentally obtained values for pressure. Note that the total force on the airfoil, and thus lift and drag, are comprised of a pressure and a viscous component. In the case of lift, the viscous contribution is entirely negligible, while in the case of drag this is not the case. Therefore, in what follows, pressure lift will be described as lift, while a distinction will be made between total and pressure drag where appropriate. Without further ado, the pressure force exerted on the airfoil is formally expressed as follows:

$$\vec{F}_{pressure} = \int_S -p\vec{n}\ \ dS = \int_S -(p - p_{ref})\vec{n}\ \ dS$$

Where $S$ is the wetted surface of the airfoil and $\vec{n}$ is its outwards facing normal. Notice that, by virtue of Gauss's divergence theorem, the introduction of any constant reference pressure does not affect the resultant force. Indeed, as previously explained, the pressures measured in this experiment are referenced to $p_2$. To obtain lift and drag from this integral, the forces are first projected in body axes and the integrals parametrized using the non-dimensional chordwise station, denoted as $\bar{x}$.

$$F_x = \int_0^1 -(p - p_2)(\bar{x})\ n_x(\bar{x})\ \ d\bar{x}$$

$$F_z = \int_0^1 -(p - p_2)(\bar{x})\ n_z(\bar{x})\ \ d\bar{x}$$

Where $x$ and $z$ are the chordwise and perpendicular directions in body axes, respectively. Using the measured values of $p - p_2$, referenced to the adequate $\bar{x}$, these integrals can be numerically computed by, for instance, the trapezoid method. Thus, the only unknown terms at this point are the two components of the outwards facing normal, which must be computed at the $\bar{x}$ stations corresponding to the pressure ports. Their computation is done according to the following procedure. First, consider two functions of $x$ and $z$ defined as follows:

$$f_{upper}(x, z) = z - z_{upper}(x)$$

$$f_{lower}(x, z) = z - z_{lower}(x)$$

Where $z_{upper}(x)$ and $z_{lower}(x)$ represent the upper and lower surfaces of the airfoil. Their analytical expressions are readily available in [1]. These functions are such that the upper and lower surfaces of the airfoil represent level curves for $f = 0$. Therefore, their gradient, evaluated at any point on the curve, must be perpendicular to it and point in the direction of maximum growth of $f$. For the upper surface, this is the outwards facing normal, while for the lower surface it is the inwards one, so a simple sign change must be applied. All in all, this method provides the following expressions for the unitary outwards facing normal at any point on the airfoil:

$$n_{x,upper} = \frac{-\frac{dz_{upper}}{dx}}{\sqrt{1 + \left(\frac{dz_{upper}}{dx}\right)^2}}$$

$$n_{z,upper} = \frac{1}{\sqrt{1 + \left(\frac{dz_{upper}}{dx}\right)^2}}$$

$$n_{x,lower} = \frac{\frac{dz_{lower}}{dx}}{\sqrt{1 + \left(\frac{dz_{lower}}{dx}\right)^2}}$$

$$n_{z,lower} = \frac{-1}{\sqrt{1 + \left(\frac{dz_{lower}}{dx}\right)^2}}$$

Using these expressions to compute the aforementioned integrals, the forces measured in body axes can be converted to lift and pressure drag by a simple rotation into wind axes:

$$L = F_x\ cos(\alpha) - F_z\ sin(\alpha)$$

$$D_{pressure} = F_x\ sin(\alpha) + F_z\ sin(\alpha)$$

Finally, these forces are converted into coefficients by using the reference surface of the airfoil and the dynamic pressure on the test chamber, in an identical manner to that shown in the previous experiment report.

Regarding the velocity measurements on the airfoil's wake, they can be used to approximate the total drag by means of a momentum balance. Indeed, the change of the momentum in the wind tunnel axis direction before and after the airfoil is opposite to the force exerted on the airfoil in said direction by the airflow. In other words, one can indirectly evaluate total drag by applying a momentum balance to a carefully selected control volume. In this case, such volume is a rectangle, two sides of which are perpendicular to the wind tunnel axis, while the remaining two are parallel to it. Such rectangle is chosen to cover the entirety of the airfoil, with room to spare, in the axial direction of the tunnel. The idea is to have the upstream inlet on the freestream, while the downstream outlet is in a station sufficiently far back to capture the fully formed wake, in which the velocity profile is measured as described in the previous section. For the remaining boundaries, an intuitive approach would be to place them on the tunnel walls. However, this would include the boundary layer that forms on them, introducing a significant error. Therefore, another approximated approach is used: the limiting boundaries are placed sufficiently far from the walls to be outside of their boundary layer, yet far enough from the wake to regain a somewhat constant velocity profile. Then, the boundaries parallel to the tunnel walls are assumed to be impenetrable, so that the momentum flow across them is neglected. This approximation is reasonable given that the airfoil, as a streamline shaped, affects only the flow in its vicinity, as long as it remains attached. Bearing this in mind, total drag can be computed by the following expression:

$$D = \int_{h_{min}}^{h_{max}} \rho U_\infty^2 - \rho U(h)^2 \ dh$$

Where the $h$ coordinate is parallel to the inlet and outlet boundaries, $\rho$ is the air's density and $U_\infty$ is the freestream velocity, determined as in the previous experiment. Thus, the first term in the integral represents the flow of momentum into the control volume through the upstream inlet, whereas the second term represents the momentum leaving the control volume through the downstream outlet, where the velocity is measured. Thus, $U(h)$ represents the measured velocity profile. Finally, this can be converted into a value for $C_d$ by dividing by the factor $\frac{1}{2}\rho U_\infty^2 c$, where $c$ is the airfoil's chord. Defining $\bar{h} = \frac{h}{c}$, the final expression is:

$$D = 2 \int_{\bar{h}_{min}}^{\bar{h}_{max}} 1 - \left(\frac{U(\bar{h})}{U_\infty}\right)^2 \ d\bar{h}$$

# 4. Experiment Results

## 4.1. Lift and Drag Coefficients

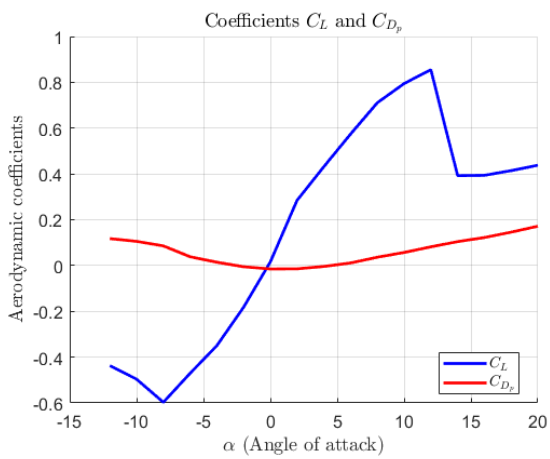


Figure 1: Variation of lift ($C_L$) and pressure drag ($C_{D_P}$) coefficients as a function of the angle of attack ($\alpha$).

Figure 1 shows the variation of the lift coefficient ($C_L$) and pressure drag coefficient ($C_{D_P}$) with respect to the angle of attack ($\alpha$). The results highlight key aerodynamic trends:

- **Lift Coefficient ($C_L$):** As $\alpha$ increases, $C_L$ follows a linear trend initially, which corresponds to the pre-stall region. A sharp decrease in $C_L$ is observed around $\alpha \approx 12°$, indicating the stall region.
- **Pressure drag Coefficient ($C_{D_P}$):** $C_D$ shows a moderate increase with $\alpha$, as expected due to increasing pressure drag. The values remain relatively small in the pre-stall regime and rise as $\alpha$ approaches stall.

## 4.2. Approximate aerodynamic polar (Lift vs Pressure drag)

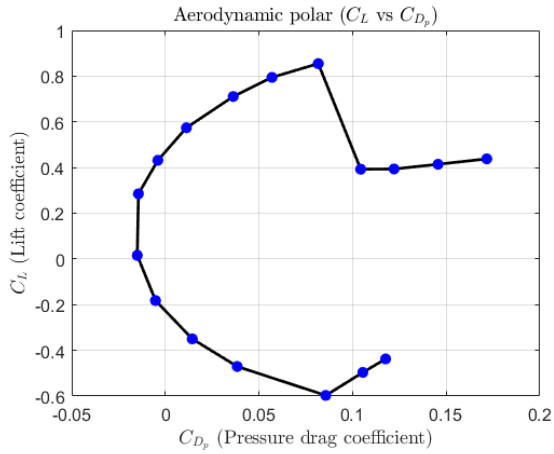


Figure 2: Approximate aerodynamic polar showing the relationship between the lift coefficient ($C_L$) and pressure drag coefficient ($C_{D_P}$).

The approximate aerodynamic polar plot, shown in Figure 2, illustrates the relationship between $C_L$ and $C_{D_P}$. Key observations include:

- **Efficiency:** The polar exhibits the expected parabolic trend, where lower drag coefficients correspond to higher lift coefficients in the pre-stall region. This is indicative of the efficient aerodynamic behavior of the airfoil.
- **Post-Stall Behavior:** After the stall point, $C_L$ decreases, while $C_{D_P}$ continues to rise, resulting in a noticeable kink in the polar curve. This highlights the loss of lift efficiency and increased drag in the post-stall regime.
- **Key Design Considerations:** The optimal lift-to-drag ratio occurs at the point of maximum curvature on the polar plot, which is crucial for aerodynamic optimization.

## 4.3. Wake Velocity Profile Analysis

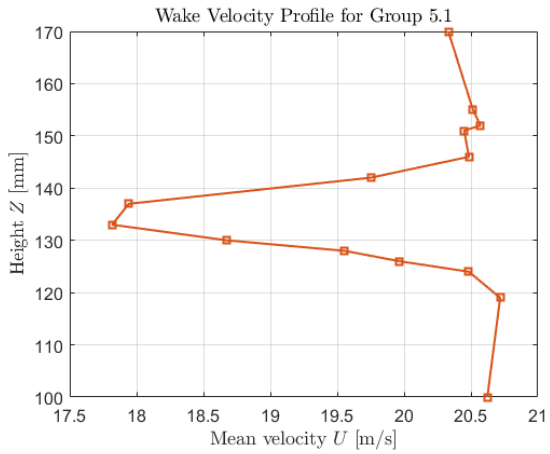


Figure 3: Wake Velocity Profile for Group 5.1. The mean velocity, $U$, is plotted as a function of height, $Z$.

The wake velocity profile, depicted in Figure 3, shows the mean velocity, $U$, as a function of height, $Z$, for $\alpha = 8°$. The data were collected using a hot-wire anemometer, a high-precision instrument widely used in fluid dynamics to measure instantaneous velocity components in a flow. Hot-wire anemometry operates based on the principle of convective heat transfer, where the rate of cooling of a heated wire depends on the local velocity of the surrounding fluid.

- The velocity distribution demonstrates significant variations across the wake. The highest velocities are observed in the regions farthest from the centerline of the wake, reaching approximately 20.5 m/s.
- Near the wake's centerline, a velocity deficit is evident, with values decreasing to around 18 m/s. This behavior is characteristic of the wake region behind an object, where the fluid experiences energy losses due to viscous effects and turbulence.

From the wake analysis, the drag coefficient, $C_D$, was calculated to be 0.0508. This value accounts for the combined effects of viscosity and turbulence. The wake profile highlights the influence of these effects, as the observed velocity deficit and fluctuations directly result from energy dissipation in the flow.

## 4.4. Final Results

The experiments conducted aimed to validate whether the NACA 23012 airfoil is optimal for achieving a lift coefficient of $C_L = 0.3$. To this end, a detailed analysis was performed based on the collected data.

Initially, the viscous drag coefficient ($C_{D\text{v}}$) was determined from wind tunnel measurements at an angle of attack $\alpha = 8°$. At this angle, the total drag coefficient was measured as $C_D = 0.0508$, which satisfies the following relation:

$$C_D = C_{D\text{v}} + C_{D\text{p}},$$

where $C_{D\text{v}}$ represents the viscous drag coefficient and $C_{D\text{p}}$ corresponds to the pressure drag coefficient. To simplify the analysis, it was assumed that $C_{D\text{v}}$, being closely related to the boundary layer, remains approximately constant across the tested conditions.

The following key results were obtained:

- Lift coefficient ($C_L$) and drag coefficient ($C_D$) variations as functions of the angle of attack ($\alpha$).
- The aerodynamic polar curve, illustrating the relationship between $C_L$ and $C_D$.

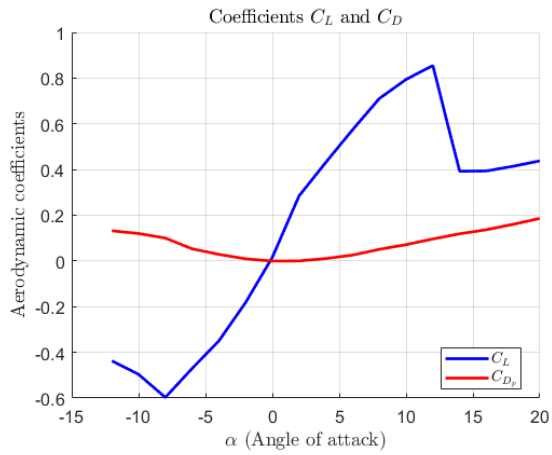


Figure 4: Lift ($C_L$) and drag ($C_D$) coefficients as functions of the angle of attack ($\alpha$).

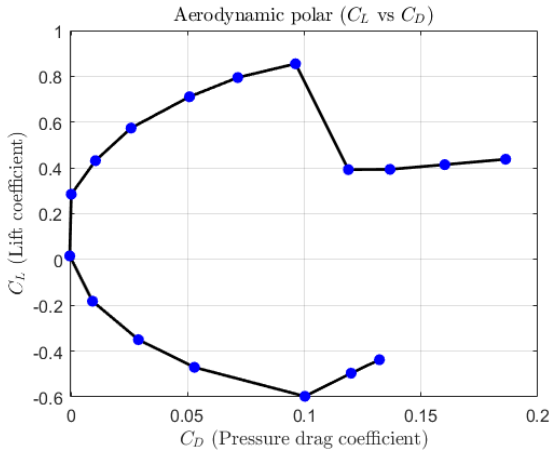


Figure 5: Aerodynamic polar curve ($C_L$ vs. $C_D$).

Additionally, the aerodynamic efficiency, expressed as the lift-to-drag ratio ($L/D$), was evaluated.

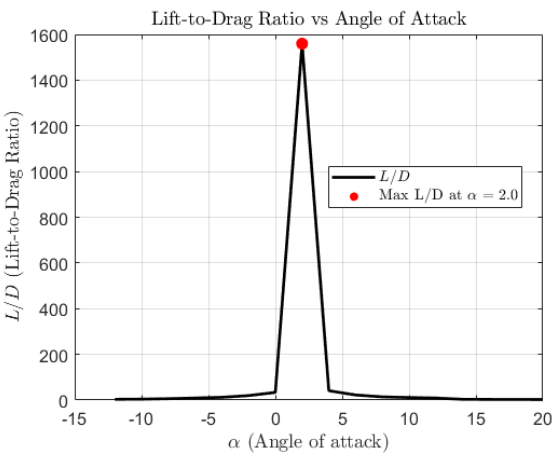


Figure 6: Lift-to-drag ratio ($L/D$) as a function of the angle of attack ($\alpha$).

Figure 6 highlights the optimal angle of attack, $\alpha = 2°$, where the lift-to-drag ratio reaches its maximum value of $L/D = 1560$. This angle also aligns with the design condition where $C_L = 0.3$, which gives confidence in the accuracy of the numerical results.

## 5. Conclusions

The experimental results obtained in this study provide valuable insights into the aerodynamic behavior of the NACA 23012 airfoil under varying conditions. The following conclusions can be drawn:

- **Lift and Drag Characteristics:** The lift coefficient ($C_L$) shows a linear increase with the angle of attack ($\alpha$) in the pre-stall region, with a maximum observed near $\alpha \approx 12°$. Beyond this point, $C_L$ decreases sharply, indicating stall. The drag coefficient ($C_D$), which was calculated based on pressure drag alone, exhibits a gradual increase with $\alpha$. However, due to the exclusion of viscous contributions, negative $C_D$ values were obtained at certain angles. This is no longer the case when we consider the viscous Cd obtained with the hot-wire experiment.
- **Wake Velocity Profile and Drag Estimation:** The wake velocity profile revealed a significant velocity deficit near the centerline of the wake, consistent with energy losses caused by viscous and turbulent effects. The calculated drag coefficient, $C_D = 0.0508$, accounts for both viscous and turbulent contributions and aligns with expectations for similar airfoil configurations.
- **Aerodynamic Polar and Optimal Design:** The aerodynamic polar confirms expected trends, with optimal lift-to-drag efficiency observed around $C_L = 0.3$. For the NACA 23012 airfoil, this corresponds to a theoretical design point near $\alpha = 2°$.

Overall, this study has shown different possibilities for airfoil lift and drag estimation based on pressure and velocity measurements, as well as the techniques used for them. Results match what was expected from NACA and aerodynamic theory, which lends creedence to the accuracy of the results.

## References

[1] Naca airfoils. `https://en.wikipedia.org/wiki/NACA_airfoil`. November 2024.